WHITEBOX

The Final Specification: One Framework That Opens Every Model to Its Provable Degree

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Abstract. This document is the complete specification for WHITEBOX: the mathematics, the software architecture, the algorithm, the models, the experiments, the schedule, and the standards of proof. The thesis is that the black box can be opened *to a measured degree*: for any trained network, a single procedure — fix the gauge, certify what survives, extract the certified structure, count the mechanisms, test whether local explanations glue, rebuild the behavior from certified parts, and measure the residue — yields an answer to each of the seven black-box problems together with a coverage gap stating exactly how much computation remains unexplained. On models small enough to check exactly, the gap is driven to zero and the opening is complete and proven. On frontier-class models the same procedure runs and the gap is reported honestly. The mathematics is gauge theory on parameter space, representation theory of finite groups, persistent and zigzag homology, spectral and differential geometry of representation manifolds, Čech cohomology of explanation presheaves, random matrix theory, and parameterized complexity. Every experiment is pre-registered with numeric pass criteria, paired with the null that would kill it, and tagged with its novelty status. This is the fourth version of the project; the first three failed by measuring before validating. This one validates first, then opens.

Contents

1 Mission and Framework **2**

1.1 The claim . . . 2
1.2 Why this is the first time . . . 2
1.3 The seven problems . . . 2
1.4 Relation to the running program . . . 2

2 Mathematical Foundations **2**

2.1 Networks, realization, fibers . . . 2
2.2 The gauge group . . . 2
2.3 Transformer gauge . . . 3
2.4 Invariance classes and the criterion . . . 3
2.5 Path products: exact gauge-invariant attributions . . . 3
2.6 Representation theory for the provable tier . . . 3
2.7 Continuous attractor networks for the provable tier . . . 3
2.8 Persistent, zigzag, and multiparameter homology . . . 4
2.9 Orbits and uniqueness across fibers . . . 4
2.10 Spectral geometry of representation manifolds . . . 4
2.11 Differential geometry: curvature and Lipschitz structure . . . 4
2.12 Čech cohomology of explanations . . . 4
2.13 Random matrix theory for the residue . . . 5
2.14 Parameterized complexity and legibility . . . 5
2.15 Invariant theory . . . 5

2.16 Statistical discipline 5

3 The `open()` Algorithm **5**

4 Library Architecture **6**

5 The Model Ladder **6**

6 The Seven Problems: Certified Procedures **7**

6.1 P0 — Is it explainable 7
6.2 P1 — What it learned 7
6.3 P2 — Why this decision 7
6.4 P3 — When bias entered 7
6.5 P4 — What it does not know 7
6.6 P5 — Where it will fail 7
6.7 P6 — Memorizing vs learning 7
6.8 P7 — How representations evolve 8

7 Reconstruction and Verification (Block R) **8**

8 Experiment Blocks **8**

8.1 Running blocks as components 8
8.2 Block O — The opener, per tier 9
8.3 Block F (running) — amendments in force 9
8.4 Blocks K, V, M, C, B, D, Z (running) 9

9 Seven-Day Schedule **10**

10 Deliverables **10**

11 Novelty Board **11**

12 Operating Constitution **11**

1 Mission and Framework

1.1 The claim

For any trained neural network f_θ and behavior B, the procedure $\texttt{open}(f_\theta, B)$ produces: (i) a certified answer to each of the seven black-box problems P1–P7 for that model and behavior, where "certified" means invariant under the network's gauge group; (ii) a reconstruction of B from certified structure alone, with a behavioral-equivalence verdict; (iii) a coverage gap $\gamma \in [0, 1]$ — the structured fraction of the unexplained residue — stating how much of the computation was not reached. The box is open to degree $1 - \gamma$. On the provable tier, $\gamma \approx 0$ and the opening is complete.

1.2 Why this is the first time

Every prior interpretability result performed extraction (step 4 below) and stopped. None certified its findings against the network's symmetries; none counted mechanisms across independent trainings; none tested whether local explanations glue; none rebuilt the behavior from its own findings; none measured what it missed. Each of those five omissions is a way a claimed "opening" can be false without anyone noticing. WHITEBOX closes all five, on every model, with one framework.

1.3 The seven problems

P1 what it learned; P2 why a given decision; P3 when and how bias entered; P4 what it does not know; P5 where it will fail; P6 memorizing versus learning; P7 how representations evolve during training. P0 (added) — is the system explainable at all — is the gate that must be passed first.

1.4 Relation to the running program

Blocks F, K, V, M, C, B, D, Z (currently executing) build and validate the instruments. This document assembles them into $\texttt{open}()$ and adds the opener blocks O (per model tier) and R (reconstruction). Nothing running is discarded; every block becomes a component (Section 8.1).

2 Mathematical Foundations

2.1 Networks, realization, fibers

A ReLU multilayer perceptron with widths $n_0, \ldots, n_L$:

$$f_\theta(x) = W_L\, \sigma\big(W_{L-1} \cdots \sigma(W_1 x + b_1) \cdots + b_{L-1}\big) + b_L, \qquad \sigma(t) = \max(t, 0).$$

For input set X, the realization map $R : \Theta \to \mathbb{R}^{|X| d_{\text{out}}}$, $R(\theta) = (f_\theta(x))_{x \in X}$. Fiber $F_y = R^{-1}(y)$. Transformers are handled analogously with attention heads as additional structured units (Section 2.3).

2.2 The gauge group

Definition 2.1 (Gauge group)**.** $G_{\mathcal{A}} = \{g : \Theta \to \Theta \mid R \circ g = R \text{ on all of } \Theta\}$.

For ReLU MLPs, $G \supseteq \prod_{\ell=1}^{L-1} \big(S_{n_\ell} \ltimes (\mathbb{R}_{>0})^{n_\ell}\big)$: permutation π_ℓ acts by $W_\ell \mapsto P_{\pi_\ell} W_\ell$, $b_\ell \mapsto P_{\pi_\ell} b_\ell$, $W_{\ell+1} \mapsto W_{\ell+1} P_{\pi_\ell}^\top$; scaling $\alpha \in (\mathbb{R}_{>0})^{n_\ell}$ acts by $W_\ell \mapsto \mathrm{diag}(\alpha) W_\ell$, $b_\ell \mapsto \mathrm{diag}(\alpha) b_\ell$, $W_{\ell+1} \mapsto W_{\ell+1} \mathrm{diag}(\alpha)^{-1}$.

Hidden symmetries. $\dim \ker dR_\theta$ (numerically, via SVD of the Jacobian of R at θ) versus $\sum_\ell n_\ell$ (the scaling dimension). Excess directions arise from dead units, duplicate units, and units linear on X; each is detected and classified. Unexplained excess is logged and the specimen flagged G-incomplete.

Proposition 2.1 (Trivial stabilizer)**.** For generic θ (pairwise distinct incoming weight vectors, no dead units), $\mathrm{Stab}_G(\theta) = \{e\}$. Consequently distinct sub-circuits of a single generic network are never gauge-equivalent.

Proof sketch. A permutation fixing θ maps each unit to a unit with identical incoming weights and bias; genericity forbids this except the identity. A scaling fixing θ satisfies $\alpha_u w_u = w_u$ with $w_u \neq 0$, so $\alpha_u = 1$. □

This proposition is why within-network orbit counting returned $K/N = 1$ identically (Block F, week of 1 Sept) and why uniqueness must be tested *across* fibers (Section 2.9).

2.3 Transformer gauge

For a head with $W_Q, W_K \in \mathbb{R}^{d\times d_h}$, $W_V \in \mathbb{R}^{d\times d_h}$, $W_O \in \mathbb{R}^{d_h\times d}$: (i) head permutations within a layer; (ii) $W_Q \mapsto W_Q A$, $W_K \mapsto W_K A^{-\top}$ for $A \in GL(d_h)$ (with RoPE, A restricted to commute with the rotation blocks); (iii) $W_V \mapsto W_V B$, $W_O \mapsto B^{-1} W_O$ for $B \in GL(d_h)$; (iv) MLP unit permutations; (v) LayerNorm/RMSNorm gain rescalings compensated downstream. GELU is not homogeneous, so MLP scalings are exact only through normalization layers; the implementation distinguishes exact from approximate symmetries and reports the latter's residual.

2.4 Invariance classes and the criterion

Definition 2.2. For $Q : \Theta \to \mathbb{R}^m$ and samples $g_1, \ldots, g_n \in G$: Q is *invariant* if $Q(g\theta) = Q(\theta)$ (CV$< 10^{-3}$); *covariant* if $Q(g\theta) = \rho(g)Q(\theta)$ for a fitted character (e.g. α^k under single-unit scaling, exponent k fitted with $R^2 > 0.99$); *artifact* otherwise.

Criterion. A statement about the internals of θ is well-defined iff it is a function on Θ/G. Covariant quantities are normalized by their character. Artifacts are discarded. This extends Implementation Invariance (Sundararajan et al. 2017) from input attributions to all internal quantities.

2.5 Path products: exact gauge-invariant attributions

For input x, let $D_\ell(x) = \mathrm{diag}(\mathbb{1}[W_\ell h_{\ell-1} + b_\ell > 0])$. Then

$$f_\theta(x) = A(x)\, x + c(x), \qquad A(x) = W_L D_{L-1}(x) W_{L-1} \cdots D_1(x) W_1.$$

Under a gauge transformation at layer ℓ, $W_\ell \mapsto PSW_\ell$, $W_{\ell+1} \mapsto W_{\ell+1} S^{-1} P^\top$, and $D_\ell \mapsto P D_\ell P^\top$ (the pattern permutes; positive scaling preserves sign), so $W_{\ell+1} D_\ell W_\ell \mapsto W_{\ell+1} S^{-1} P^\top P D_\ell P^\top P S W_\ell = W_{\ell+1} D_\ell W_\ell$. Hence $A(x)$ and $c(x)$ are exactly invariant. $A(x)$ is the model's computation on x; its entries are certified per-decision attributions (P2). Integrated gradients agree with $A(x)x$ on regions of constant activation pattern; this is the D1 anchor.

2.6 Representation theory for the provable tier

Modular addition $(a, b) \mapsto a + b \bmod p$. Characters of $\mathbb{Z}_p$: $\chi_k(x) = e^{2\pi i k x/p}$, $k = 0, \ldots, p-1$; real irreps are the 2-dimensional rotation blocks $\{\cos(2\pi kx/p), \sin(2\pi kx/p)\}$ for $k \neq 0$. Nanda et al. and Chughtai et al. established that grokked transformers implement: embedding $E \in \mathbb{R}^{p\times d}$ concentrated on a small set $\mathcal{K}$ of frequencies; attention/MLP computing products $\cos(\omega_k a)\cos(\omega_k b)$ etc.; unembedding as inverse Fourier read-off via $\cos(\omega_k(a + b - c))$.

Certified extraction. Compute the discrete Fourier transform of E along the token axis: $\hat{E}_k = \sum_x E_x e^{-2\pi i k x/p}$. The frequency set $\mathcal{K} = \{k : \|\hat{E}_k\|^2 / \sum_j \|\hat{E}_j\|^2 > \tau\}$ is invariant under all of G (permuting or rescaling hidden units does not touch the token axis; rotations of embedding space preserve norms). The *basis* within each irrep is gauge. Fourier concentration $\kappa = \sum_{k\in\mathcal{K}} \|\hat{E}_k\|^2 / \sum_k \|\hat{E}_k\|^2$ is invariant. Per-frequency weights and the neuron-level trig-product structure are extracted and passed through the gate.

2.7 Continuous attractor networks for the provable tier

Ring attractor RNN: $\tau \dot{x} = -x + W\sigma(x) + I(t)$ with $W_{ij} = J_0 + J_1 \cos(\phi_i - \phi_j)$ (or trained from scratch on angular integration). The attractor manifold is $\mathcal{M} \cong S^1$. Certified extraction: sample activity $\{x(t)\}$, compute H_1 of the Vietoris–

Rips complex ($\beta_1 = 1$ persistent bar), the angular coordinate via the circular coordinate of the persistent cocycle (de Silva–Morozov–Vejdemo-Johansson), and the tangent dynamics on $\mathcal{M}$. Unit permutations are isometries (invariant); per-unit scalings are not, so activity is normalized per unit before the topological computation (this normalization is itself a gauge-fixing and is recorded).

2.8 Persistent, zigzag, and multiparameter homology

Vietoris–Rips filtration on activation clouds; diagrams D_k; W_p distances; stability. Properties used: isometry invariance (permutations), degree-1 scaling covariance (uniform scaling), and *non*-covariance under per-unit scaling (measured). **Normalized persistence** (Z5): divide all diagram coordinates by the median pairwise distance of the cloud; the result is degree-0 under uniform scaling. Zigzag persistence over checkpoints $A_0 \hookleftarrow A_0 \cup A_1 \hookrightarrow A_1 \hookleftarrow \cdots$ tracks individual classes in training time. Two-parameter persistence (scale $\times$ training fraction) via rank invariant / Hilbert function.

2.9 Orbits and uniqueness across fibers

Train $\theta_1, \ldots, \theta_s$ to the same target. Extract minimal certified circuits C_j. $C_i \sim C_j$ iff $\exists g \in G$ with $g \cdot (\theta_i|_{C_i}) \approx \theta_j|_{C_j}$ (weights on the support match to tolerance 10^{-4} after quotienting hidden directions). Alignment is computed by permutation matching (Hungarian on unit-weight distances) followed by per-unit scaling fit. The number of classes estimates $\pi_0(\mathcal{C}(F)/G)$. Z4 tests path-connectedness within classes by interpolating in parameter space and checking validity along the path.

Conjecture 2.1 (Uniqueness up to gauge)**.** For a target y and architecture class, $\pi_0(\mathcal{C}(F_y)/G)$ is finite and equals the number of distinct named algorithms realizable at that width.

2.10 Spectral geometry of representation manifolds

Activation cloud $A = \{h(x)\} \subset \mathbb{R}^n$. k-NN graph with Gaussian weights $w_{ij} = \exp(-\|h_i - h_j\|^2/\epsilon)$; normalized Laplacian $\mathcal{L} = I - D^{-1/2} W D^{-1/2}$; eigenpairs (λ_i, ϕ_i). Belkin–Niyogi: $\mathcal{L} \to$ Laplace–Beltrami on $\mathcal{M}$ as $n \to \infty, \epsilon \to 0$. Diffusion coordinates $\Psi_t(h) = (\lambda_i^t \phi_i(h))_i$; diffusion distance $d_t(h, h') = \|\Psi_t(h) - \Psi_t(h')\|$. **Spectral gap** $\lambda_2 - \lambda_1$ and the number of eigenvalues below a threshold estimate the manifold's cluster/component structure; the eigenvalue flow across training checkpoints (P7) records when structure crystallizes. **Out-of-manifold distance** (P4): Nyström extension of ϕ_i to a new input h^*; $d_{\mathcal{M}}(h^*) = \min_{h \in A} d_t(h^*, h)$; inputs with $d_{\mathcal{M}}$ above the in-distribution 99th percentile are "not known." Invariance: the graph is built from pairwise distances, so permutation-invariant; per-unit scaling requires normalization (recorded as gauge-fixing).

2.11 Differential geometry: curvature and Lipschitz structure

Local tangent space $T_h\mathcal{M}$ via PCA on the k-NN neighborhood; normal component of neighbors fitted by a quadratic form Q_h (second fundamental form estimate); principal curvatures = eigenvalues of Q_h. Local Lipschitz constant of the certified map: $\mathrm{Lip}(h) = \|A(x)\|_2$ (spectral norm of the path product) and its variation $\|A(x) - A(x')\|_2 / \|x - x'\|$ over neighbors. **P5 predictor:** $\rho(x) = \|Q_{h(x)}\| \cdot \mathrm{Lip}(x)$. Prediction: failure probability under perturbation of radius r is monotone in ρ. Tested by actual perturbation.

2.12 Čech cohomology of explanations

Cover inputs by overlapping regions $\{U_i\}$; local certified explanations E_i; restrictions $E_i|_{ij}$ by re-scoring on $U_i \cap U_j$; agreement iff Jaccard of thresholded edge sets $\geq \tau$. H^0 = consistent explanation families; H^1 = obstruction cocycles (cycles of overlaps with unresolvable disagreement). Framework due to arXiv 2603.17141 and the gauge/sheaf-atlas paper (cited). We compute the numbers and localize mechanism boundaries. Sanity gate: a known single-mechanism task must return $H^1 \approx 0$.

2.13 Random matrix theory for the residue

Residual $r_t = h_t - \hat{h}_t$ where $\hat{h}_t$ is the reconstruction from certified structure (Section 3, step 7). $\Sigma = \frac{1}{n}\sum r_t r_t^\top$. Under isotropic noise of variance σ^2 with $d/n \to \gamma$, the spectrum follows Marchenko–Pastur on $[\sigma^2(1-\sqrt{\gamma})^2, \sigma^2(1+\sqrt{\gamma})^2]$; the top eigenvalue fluctuates on the Tracy–Widom scale. **Coverage gap** $\gamma_{\text{cov}} = \sum_{\lambda_i > \lambda_+} \lambda_i / \operatorname{tr}\Sigma$ (structured fraction of the residue). Nulls: synthetic Gaussian residue with matched marginals; coordinate-shuffled residue.

2.14 Parameterized complexity and legibility

Circuit discovery is NP-hard, inapproximable, and globally Σ_2^p-complete (Adolfi et al. 2025). Parameters on the attribution graph Γ_θ: modularity Q (Leiden), treewidth upper bound (min-fill), sparsity. **P0 gate:** `open()` reports the legibility profile and refuses full extraction if the model lies in the empirically intractable regime of the V-block phase diagram, returning a partial scorecard instead.

Conjecture 2.2 (Legibility)**.** Circuit discovery restricted to attribution graphs of treewidth $\leq k$ is fixed-parameter tractable in k; trained networks concentrate at small k.

2.15 Invariant theory

The ring $\mathbb{R}[\Theta]^G$ for $G = \prod S_{n_\ell} \ltimes (\mathbb{R}_{>0})^{n_\ell}$. Scaling-invariant monomials are products along input–output paths (telescoping); permutation-symmetrizing gives candidate generators. Whether the path-product algebra generates $\mathbb{R}[\Theta]^G$ is an open question posed to JVHM. The empirical completeness test (predict behavior from dictionary entries; artifacts must add nothing) is the operational stand-in.

2.16 Statistical discipline

Pre-registered thresholds; feature-shuffle, label-permutation, MP, and planted-structure nulls; bootstrap CIs; effect sizes with p-values; kill attempts reported. Baseline or no claim.

3 The `open()` Algorithm

Input: model f_θ; behavior B (task, input distribution $\mathcal{D}_B$, metric); optional seeds $\theta_1, \ldots, \theta_s$ for the same B; optional training checkpoints.
Output: scorecard $\{P0, \ldots, P7\}$ with certified answers, reconstruction verdict, coverage gap γ_{cov}, provenance (which methods produced which certified objects, with nulls).

1. **Gauge.** Compute G (permutations, scalings, hidden via Jacobian corank). Record exact vs approximate symmetries.
2. **Gate (P0).** Attribution graph; Q, treewidth bound, sparsity; legibility verdict from the V phase diagram. If intractable: report and proceed with partial extraction only.
3. **Extract.** Run every available method: path products (P2), Fourier/irrep decomposition where a group acts (P1), persistent H_1 and circular coordinates for attractor structure (P1), SAE/probe features (P1), spectral-geometric embedding (P4), curvature/Lipschitz field (P5), Fourier concentration and group-structure criterion (P6), zigzag/spectral flow over checkpoints if present (P3, P7).
4. **Certify.** Every candidate through `check_invariance`. Discard artifacts; normalize covariants; keep invariants. Log the audit.
5. **Count.** If seeds are provided: cross-fiber circuit alignment; report $|\pi_0|$ estimate and the named algorithms.
6. **Glue.** Cover $\mathcal{D}_B$; local certified explanations; H^0, H^1; mechanism boundaries.
7. **Reconstruct.** Build $\hat{f}$ from certified objects only (no weight copying): irreps + trig-product layer + inverse Fourier readout for group tasks; circle + tangent dynamics for attractors; certified circuits for general tasks. Compute the reconstruction $\hat{h}_t$ of activations. Residual r_t; coverage gap γ_{cov} via MP.

8. **Verify.** Behavioral equivalence of $\hat{f}$ vs f_θ: accuracy; logit correlation; perturbation-robustness curves (Gaussian and adversarial radii); out-of-distribution agreement; per-object ablation symmetry (ablating object k in both must have the same effect). Pass iff all pre-registered tolerances met.
9. **Report.** Scorecard; degree of opening $1 - \gamma_{\text{cov}}$; if verify fails, the residue's top spectral directions are returned as targets for the next extraction round.

4 Library Architecture

```
whiteboxtopo/
  symmetry/ permutations(model) scalings(model) hidden(model, X) apply(model, g)
            transformer_gauge(model) sample(model, n, kind="perm"|"scale"|"both")
  invariants/ check_invariance(metric_fn, model, n=100, align=False) -> Report
            path_product(model, x) -> (A, c) # exact invariant attribution
            fourier_irreps(embedding, p, tau) -> (K, kappa, weights)
            circular_coordinate(cloud) -> theta # persistent cocycle
  nulls/ feature_shuffle(X) label_permute(y) marchenko_pastur(spec, d, n)
            planted(kind, size) tracy_widom_edge(d, n, sigma2, p)
  graphs/ attribution_graph(model, X, thr) modularity(G) treewidth_ub(G) sparsity(G)
  discovery/ find_circuit(model, task, faith=0.9, seed) -> Circuit # frozen pipeline
  topology/ vr_diagrams(cloud) normalized_persistence(cloud) zigzag(clouds)
            multiparam(cloud, params) mapper(cloud, lens) graph_ph(G)
  spectral/ laplacian(cloud, k, eps) diffusion_coords(L, t) nystrom(L, h_new)
            out_of_manifold(cloud, h_new) eigenflow(checkpoint_clouds)
  geometry/ tangent_pca(cloud, k) curvature(cloud) lipschitz_field(model, X)
            failure_predictor(model, X) -> rho
  sheaf/ cover(X, k, overlap) local_explanations(model, cover) restrict(E, U)
            cech_H0_H1(cover, E, tau) -> (H0, H1, boundaries)
  align/ match_circuits(theta_i, C_i, theta_j, C_j) -> (g, dist)
            count_mechanisms(seeds, circuits) -> (K, classes)
  reconstruct/ from_irreps(K, weights, p) from_attractor(theta_coord, dynamics)
            from_circuits(certified) residual(model, recon, X) coverage_gap(residual)
  verify/ equivalence(f, f_hat, X, radii, ood) -> Verdict
            ablation_symmetry(f, f_hat, objects) -> table
  signals/ memorization_signal(model, family) legibility_score(model)
            residual_structure(sae, X) explanation_obstruction(model, X, task)
  open/ open(model, behavior, seeds=None, checkpoints=None) -> Scorecard
            scorecard.to_grid() scorecard.to_markdown()
```

Every public function: docstring, unit test, VALIDATION.md entry, null it was tested against. `CLAIMS.md` maps claim → experiment → evidence → null → failure mode → novelty status.

5 The Model Ladder

Tier	Models	What is provable	Compute
T1 Provable	Grokked 1-layer transformer on $\mathbb{Z}_{113}$ addition (clean; planted-shortcut; memorizing control), 5 seeds each. Ring-attractor RNN (trained, 128 units, angular integration), 10 seeds.	All seven answered exactly; reconstruction passes; $\gamma_{\text{cov}} \approx 0$. Complete opening.	Laptop + 5070
T2 Small real	GPT-2-small: IOI, induction, greater-than.	Certified circuits; mechanism count; H^1 boundaries; γ_{cov} measured (expected substantial).	5070
T3 Checkpoint	Pythia-410M and 1B (154 public checkpoints).	P7 on a real training run; P3 on naturally-occurring bias probes; anyone can verify.	HPCC
T4 Frontier	Llama-3-8B, Gemma-2-9B (+ Gemma Scope SAEs).	Full audit; legibility verdict; γ_{cov}; degree of opening stated.	HPCC

The headline figure is the **grid**: models (rows) $\times$ P0–P7 (columns), each cell a certified answer or "not reached," final column $1 - \gamma_{\text{cov}}$.

6 The Seven Problems: Certified Procedures

6.1 P0 — Is it explainable

Attribution graph $\to (Q, \text{tw}, \text{sp}) \to$ placement on the V phase diagram $\to$ verdict {tractable, marginal, intractable}. Null: degree-preserving rewiring. Pass: T1/T2 tractable.

6.2 P1 — What it learned

T1 group task: $\mathcal{K}$, κ, per-frequency weights, neuron trig-product assignments; certified. Pass: $\kappa > 0.9$; $\mathcal{K}$ identical across all 300 gauge samples; reconstruction from $\mathcal{K}$ alone passes verify. **T1 ring:** $\beta_1 = 1$ persistent bar with lifetime $> 5\times$ next; circular coordinate correlates with true angle $r > 0.99$; certified after per-unit normalization. **T2–T4:** certified SAE/probe feature inventory (C block), Mapper skeleton of activation space, spectral embedding; report certified fraction. Null: feature-shuffle; random-direction features.

6.3 P2 — Why this decision

Path product $A(x)$; top input contributions $A(x)_{\cdot j} x_j$; exact invariance verified (CV$< 10^{-6}$). Anchor: agreement with integrated gradients on constant-pattern regions. For transformers: attention-pattern-conditional path products (pattern frozen per input), reported with the QK caveat. Pass: invariance exact; anchor agreement $r > 0.95$.

6.4 P3 — When bias entered

T1 planted: shortcut feature (residue-class token leaking the answer) with known onset in the data; zigzag over checkpoints locates birth of the shortcut-aligned H_1/feature; Laplacian eigenflow shows the associated eigenvalue's appearance; certified. Pass: detected onset within 5% of training of the ground-truth crystallization (defined as the checkpoint where shortcut-ablation first changes accuracy by > 5 points). Null: same pipeline on the clean model must detect nothing. **T3:** naturally-occurring gender/occupation probes across Pythia checkpoints; bias-direction invariance (B2); power-adequate n (B3). Report honestly; v1's precedence claim remains retired unless T1-planted and T3 both pass.

6.5 P4 — What it does not know

Diffusion embedding of in-distribution activations; Nyström extension; $d_{\mathcal{M}}$ for held-out, OOD, and adversarial inputs. Pass: AUC> 0.9 separating in- vs out-of-distribution on T1/T2, beating softmax confidence (the baseline that beat v1). Additionally the tool-side P4: γ_{cov} of any decomposition (K block). Null: random projection embeddings.

6.6 P5 — Where it will fail

$\rho(x) = \|Q_{h(x)}\| \cdot \text{Lip}(x)$; perturb each x at radii $r \in \{0.01, \ldots, 0.5\}$; failure iff prediction flips. Pass: Spearman $\rho(\rho(x), \text{failure radius}) < -0.5$, $p < 0.01$, beating softmax-margin baseline. v1's P5 was null with a weak predictor; this is a geometric predictor with a mechanism. Null: shuffled ρ.

6.7 P6 — Memorizing vs learning

T1: Fourier concentration κ (generalizer > 0.9; memorizing control < 0.3) — a representation-theoretic criterion. **T2+:** audited `memorization_signal` per family (M block) with baselines (susceptibility, loss-dynamics detectors). Pass: κ separates with AUC 1.0 on T1; at T2+ report honestly against baselines.

6.8 P7 — How representations evolve

Zigzag intervals of H_1 classes over training; Laplacian eigenflow; normalized-persistence velocity (the corrected v1 SV); legibility drift $Q(t)$. Pass on T1: irrep appearance times form the grokking transition, spectral gap opens within 5% of the accuracy jump, and the ordering of frequencies is consistent across $\geq 4/5$ seeds. T3: same on Pythia, reported.

7 Reconstruction and Verification (Block R)

R1 — Rebuild from irreps (T1 group) *5070*

Construct $\hat{f}$: embedding rows $= \big(\cos(\omega_k a), \sin(\omega_k a)\big)_{k\in\mathcal{K}}$ with certified weights; hidden layer computing $\cos(\omega_k(a+b))$ via trig products; readout $\cos(\omega_k(a+b-c))$ summed over $\mathcal{K}$. No weights copied. Pass: accuracy $\geq f_\theta - 0.5\%$; logit correlation > 0.98.

R2 — Rebuild from attractor (T1 ring) *5070*

Construct $\hat{f}$ from the circular coordinate and fitted tangent dynamics only. Pass: integration error within 10% of original over 100-step trajectories; same drift statistics.

R3 — Rebuild from certified circuits (T2) *5070*

$\hat{f}$ = certified circuit with complement mean-ablated. Pass: faithfulness ≥ 0.9; report γ_{cov}.

R4 — Residual and coverage gap *5070/HPCC*

$r_t = h_t - \hat{h}_t$; MP null; γ_{cov} per model. Pass on T1: $\gamma_{\text{cov}} < 0.05$ with synthetic-noise null < 0.02. T2–T4: report.

R5 — Behavioral equivalence battery *5070*

Accuracy; logit correlation; Gaussian-perturbation curves at 6 radii; adversarial (PGD) robustness curves; OOD agreement (moduli/ranges unseen). Pass: all curves within pre-registered bands.

R6 — Ablation symmetry *5070*

Ablate each certified object in f_θ and its counterpart in $\hat{f}$; effect vectors must correlate $r > 0.9$. Any object failing is mis-mapped.

R7 — Negative control *5070*

Rebuild from *artifact-class* objects only; must fail (accuracy near chance). Rebuild from a random subset of certified objects; must degrade gracefully in proportion to κ removed.

R8 — Residue-guided second pass *5070*

If R5 fails, take top MP-outlier directions of r as new extraction targets; iterate steps 3–8 once; report γ_{cov} before/after.

8 Experiment Blocks

8.1 Running blocks as components

Block	What it validates	Role in `open()`
F	Gauge group; audit table; cross-fiber uniqueness	Steps 1, 4, 5
K	MP/topology nulls on residue	Step 7 (coverage gap); tool-side P4
V	Legibility phase diagram; drift	Step 2 (P0); P7
Z	Zigzag, multiparameter, graph-PH, normalized persistence	P3, P7 instruments
M, C, B	Audited P6, P1, P3 signals with baselines	Certified inputs to those cells
D	Path-product vs integrated-gradients anchor	P2 anchor

8.2 Block O — The opener, per tier

O1.1 — T1 specimens *Laptop/5070*

Train grokked $\mathbb{Z}_{113}$ transformers (clean, planted-shortcut, memorizing), 5 seeds; save checkpoints every 1%. Train ring RNNs, 10 seeds, checkpoints every 2%. Verify grokking (test acc $> 99\%$) and Fourier structure present (Nanda diagnostics). Specimens failing diagnostics are excluded and logged.

O1.2 — T1 full open() *5070*

Run steps 1–9 on every T1 specimen. Deliver: per-specimen scorecard; cross-seed mechanism count; H^0, H^1 (expect $H^1 = 0$ for clean modular addition — this is also the sheaf sanity gate); reconstruction verdict; γ_{cov}. Pass: all seven cells certified on $\geq 4/5$ clean seeds; R1/R2/R5 pass; $\gamma_{\text{cov}} < 0.05$.

O1.3 — T1 planted-bias detection *5070*

P3 procedure on planted-shortcut specimens vs clean. Pass: onset within 5%; zero false detections on clean.

O1.4 — T1 P4/P5 geometry *5070*

Diffusion OOD distance and curvature–Lipschitz failure predictor with softmax baselines. Pass thresholds per Section 6.

O2.1 — T2 GPT-2 open() *5070*

IOI, induction, greater-than. Steps 1–9 with certified circuits; H^0, H^1 with boundary localization (IOI expected $H^1 > 0$, induction ≈ 0); R3/R4. Report γ_{cov} per task.

O2.2 — T2 audit at real scale *5070*

F10 candidate list on GPT-2 (this is F11); classifications carried into the grid.

O3.1 — T3 Pythia P7/P3 *HPCC*

Eigenflow, normalized-persistence velocity, legibility drift, zigzag on layer-mid activations across all checkpoints; bias probes with B2/B3 procedures. Report.

O4.1 — T4 frontier open() *HPCC*

Llama-3-8B / Gemma-2-9B: gauge (Llama backend verified), P0, audit (F12), certified SAE inventory (C4), γ_{cov} with Gemma Scope reconstructions (K8), legibility verdict. Partial extraction; report degree of opening.

O5 — The grid *Laptop*

Assemble all scorecards into the models $\times$ P0–P7 grid with $1 - \gamma_{\text{cov}}$; one figure; one table; per-cell provenance links to evidence files.

8.3 Block F (running) — amendments in force

Cross-fiber uniqueness replaces within-network orbits as the primary F4/F5 metric; within-network $K/N = 1$ is retained as the redundancy finding (Proposition 2.1); planted kill for cross-fiber: two seeds that are literal relabelings must align ($K = 1$); seeds trained to different named algorithms must not ($K = 2$).

8.4 Blocks K, V, M, C, B, D, Z (running)

As specified in the Agent Team Handbook; outputs feed the cells above. K's MP null is reused verbatim for R4. V's diagram provides P0's verdict regions. Z5's normalized persistence replaces raw TC everywhere.

9 Seven-Day Schedule

Day	Work	Pause
1	O1.1 training (all T1 specimens, all checkpoints); `spectral/`, `geometry/`, `reconstruct/`, `verify/` modules written with unit tests; F cross-fiber amendment running.	—
2	O1.2 steps 1–6 on T1 (gauge, P0, extraction, certification, count, glue); P1 and P2 cells.	PAUSE A: T1 extraction + audit
3	O1.4 (P4, P5 geometry with baselines); P6 (κ criterion) on T1.	—
4	O1.3 (P3 planted) and P7 (zigzag + eigenflow) on T1 checkpoints.	PAUSE B: T1 all seven cells
5	R1–R7 on T1; γ_{cov}; R8 if needed.	PAUSE C: T1 opened / not
6	O2.1–O2.2 on GPT-2; HPCC jobs O3.1, O4.1 launched with checkpointing.	—
7	O5 grid; scorecards; `open()` end-to-end wrapper; paper figures; CLAIMS.md; novelty research of every claim.	PAUSE D: release review

HPCC results (O3, O4) land when they land; the grid ships with those rows marked pending and is updated.

10 Deliverables

1. **The grid** — models $\times$ P0–P7 $\times$ degree of opening. The headline figure.
2. **T1 complete-opening proof** — scorecards, reconstruction verdicts, $\gamma_{\text{cov}} \approx 0$, with the mathematical objects (irreps and weights; circle and dynamics) written out.
3. **The audit table** — which interpretability quantities are real, at toy, GPT-2, and 8B scale.
4. **The library** — `whiteboxtopo` with `open()`, all modules, tests, VALIDATION.md, CLAIMS.md, NOVELTY_BOARD.md.
5. **The paper** — "WHITEBOX: Opening Neural Networks to Their Provable Degree" — sections: gauge and certification; the algorithm; T1 complete opening; T2–T4 measured opening; the grid; what remains dark and why.

11 Novelty Board

Claim	Neighbors	Status
Mechanistic invariance audit	Implementation Invariance (2017), input-level only	Open (extension)
Cross-fiber circuit uniqueness up to gauge	Convergent learning; re-basin (whole-network permutation)	Open (circuit-level, with scalings)
Within-network multiplicity is redundancy, not gauge	Méloux 2025	Open (clarification)
Path products as exact invariant attribution	Locally-linear ReLU analysis; IG	Flagged (cite; contribution is certification framing)
Certified irrep extraction + reconstruction	Nanda; Chughtai (read-only)	Open (certify + rebuild + verify)
Ring-attractor certified extraction + rebuild	Fiete; de Silva circular coordinates	Flagged (cite; contribution is the loop)
Diffusion-distance OOD from certified manifold	Diffusion maps; OOD detection literature	Flagged
Curvature $\times$ Lipschitz failure predictor	Robustness/curvature literature	Flagged
Planted-bias detection via zigzag + eigenflow	Bias-dynamics literature	Open (planted ground truth)
MP null on residue; coverage gap	MP on weights (2025); Engels dark matter	Open (application)
Čech obstruction measured on real model	arXiv 2603.17141; sheaf atlas	Open (measurement; framework cited)
Quantitative legibility diagram	Golechha; Adolfi	Open
The closed loop and the grid	—	Open

Every claim is re-searched within 14 days of write-up; hits are cited and the claim repositioned.

12 Operating Constitution

1. SPEC.md before compute; AMENDMENTS.md for changes; thresholds never edited after results.
2. Kill before believe; report kills either way.
3. Baseline or no claim (softmax confidence/margin for P4/P5; susceptibility and loss-dynamics detectors for P6; integrated gradients for P2).
4. Negatives are deliverables.
5. Novelty status is data; TAKEN runs as anchor only.
6. Public data only in this sprint; lab data untouched.
7. Result beats framing.
8. Every shipped function carries its failure boundary.
9. The word "complete" is used only where $\gamma_{\text{cov}} < 0.05$ and R5–R7 pass. Everywhere else: "opened to degree $1 - \gamma_{\text{cov}}$."

This is the fourth version. The first measured variance and called it topology; the second found the sign flips; the third built a tautology. Each was caught by the discipline above, and each correction is now a component of the machine that opens the box. What that machine opens completely, it proves. What it cannot reach, it measures. That is the difference between this and every claim before it.